



The Gym Booking App

Your Workout, Your Schedule – Book Your Gym Time Effortlessly!

Goal

In 2023, as part of my studies at Laurea University of Applied Sciences, I developed the Gym Booking App as part of the “Defining and Designing a Software Product” course. The primary objective of this project was to design and develop a user-centered application prototype, while gaining a comprehensive understanding of Agile methodologies, specifically Scrum. The project provided a balance between both the technical and design aspects of software development, allowing me to gain hands-on experience in transforming an initial concept into a functional prototype.

Throughout the project, my main motivation was to deepen my understanding of the entire app development process, utilizing essential tools such as Miro, Figma, and Atlassian Jira and Confluence. I focused on enhancing my skills in software documentation, reporting, and prototyping, while collaborating with a cross-functional team.

Additionally, I aimed to apply software engineering principles and Agile practices to ensure the efficient delivery of a high-quality product. I actively contributed to the

design and functionality of the Gym Booking App, with a specific focus on usability, interface design, and overall user experience. I also contributed to documenting software requirements to provide a clear foundation for development. This project provided me with valuable insights into the software development lifecycle and further developed my ability to work effectively within a team, while meeting both technical and user-centric objectives.

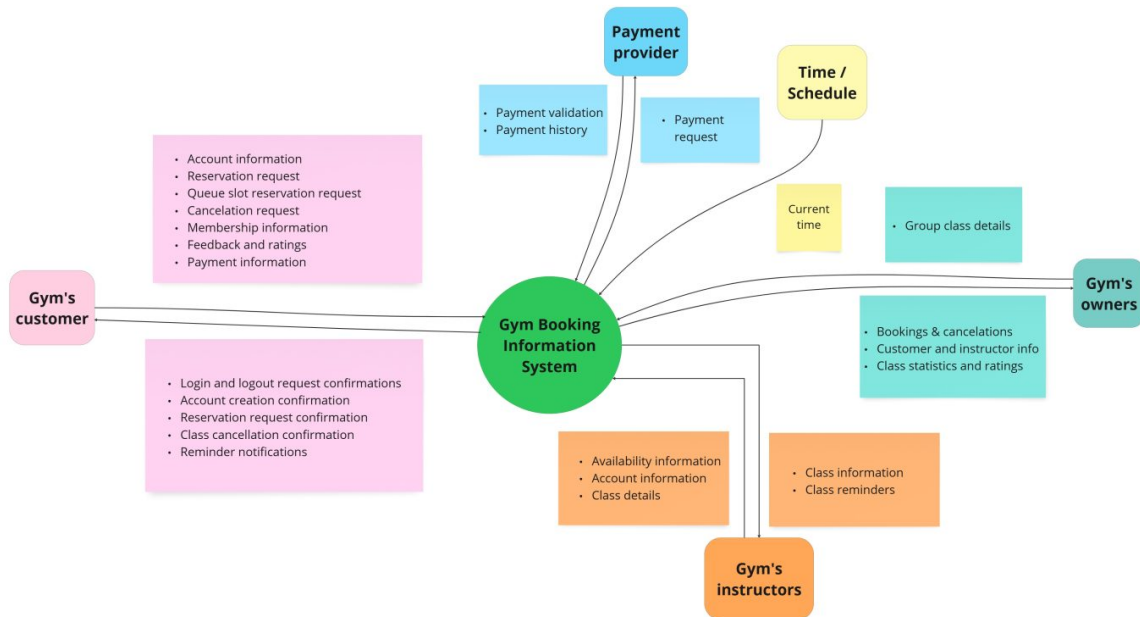
Planning and Development Process

As part of the project, I contributed to the planning and design of the Gym Booking App. My role involved creating technical documentation, designing the system architecture, and developing the app's sketches, wireframes and prototype. I created a context diagram to provide an overview of the software system, as well as data flow diagrams (DFD) to illustrate how data would flow through the app's processes. Additionally, I defined the app's APIs, software architecture, and database schemas, ensuring the system was scalable, efficient, and aligned with user needs.

1. The App's Context Diagram

I started planning by creating the app's Context Diagram to provide a high-level overview of the system, illustrating its key components and how they interact with external entities. This diagram helped to define the scope of the project, ensuring that all relevant processes, users, and external systems were clearly identified and understood. It served as a foundation for further design and development, allowing for a clearer vision of the app's structure and functionality.

Context Diagram



2. The App's APIs

The next step was defining the Gym Booking App's APIs to ensure seamless communication between the app's frontend and backend. I defined both internal and external APIs, including the **Calendar API** for organizing group class schedules, the **Sign-in/Log-in APIs** for easy access across platforms (iOS, mobile, desktop), and the **Paytrail Payment API**, which supports multiple payment options from Finnish banks like Nordea, OP, and Danske. These APIs optimized functionality, ensured smooth data flow, and laid the foundation for scalability and future integrations.

3. The Data Flow Diagram (DFD) and APIs

In the next step, I developed The Gym Booking App's Data Flow Diagram (DFD) and APIs to ensure a clear understanding of how data would flow through the system and establish seamless communication between the app's components from the perspectives of the gym owner, instructor, and user. The DFD helped map key processes such as class creation, booking, availability checking, payment processing, and feedback collection, providing insight into how data moves between the different system components. Meanwhile, developing the APIs was essential for enabling smooth interactions between the frontend and backend, ensuring efficient communication across all user types, and facilitating future scalability and integrations.

4. The Software (SW) Architecture

I defined the software (SW) architecture for The Gym Booking App using the Model-View-Controller (MVC) design pattern to ensure a clean, scalable, and maintainable structure. The app includes six distinct MVC patterns, which help separate concerns and improve the organization of the codebase.

The **Login page** shares the same view across all user roles, providing a consistent and seamless login experience. The **Login control** is linked to **Account control**, **Membership control**, and **Group class and instructor control**. This modular approach allows for easy updates and maintenance, as each control is responsible for a specific set of functions.

Group class and instructor control is further linked to **Feedback control** and **Reservation and queue control**, ensuring smooth transitions between user actions such as booking classes, leaving feedback, and managing reservations.

The app uses **MySQL** for its database, providing a reliable and scalable storage solution. The **Apache HTTP Server** serves as the web server, ensuring stable and efficient communication between the app's client and server. By implementing this architecture, I ensured that the app is scalable, maintainable, and easily adaptable to future changes and feature additions, allowing for a seamless user experience across all roles.

5. The Database Schema

I designed this database schema to ensure a structured, efficient, and scalable foundation for a gym management application. It maintains data integrity, optimizes retrieval, and establishes clear relationships between key entities.

The User entity stores login credentials, while Customers, Instructors, and Owners have personal details. Gyms are linked to instructors and owners through many-to-many relationships. Group classes include essential details like name, description, date, time, and location, with schedules managed separately. Memberships define pricing and duration, Payments track transactions, Reservations handle class bookings, and Feedback stores customer reviews.

With this schema, I created a solid framework to streamline gym operations, memberships, scheduling, and customer interactions effectively.

6. The App's Page View Sketch

I designed the app with 15 main pages, tailored to three user roles: customer, owner, and instructor. The interface and functionality adjust based on the user's role.

All users can sign in, manage their profile, customize settings, and access the **Menu and Group Class** pages. Customers can book and cancel classes, track reservations, view activity history, and give feedback. Owners can create and delete classes, while instructors can edit them.

Customers can manage memberships and payments through the **Membership, Shop, and Payment** pages, with transactions handled via Paytrail. Owners can also manage the shop and access **Gym and Statistics** pages to edit gym details and analyze customer activity.

This structured design ensures an intuitive and efficient user experience for all roles.

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Automatic Zoom

7. The App's Wireframes

Creating wireframes was the next essential step after defining the database schema because it helped me visualize the app's structure, user flow, and interactions.

Wireframes allowed me to map out each page's layout, ensuring a clear and intuitive user experience tailored to customers, owners, and instructors. By sketching the app's interface, I could identify potential usability issues early, refine navigation, and align the design with the app's functionality. This step made it easier to plan development, improve user interaction, and ensure a smooth transition from concept to implementation.

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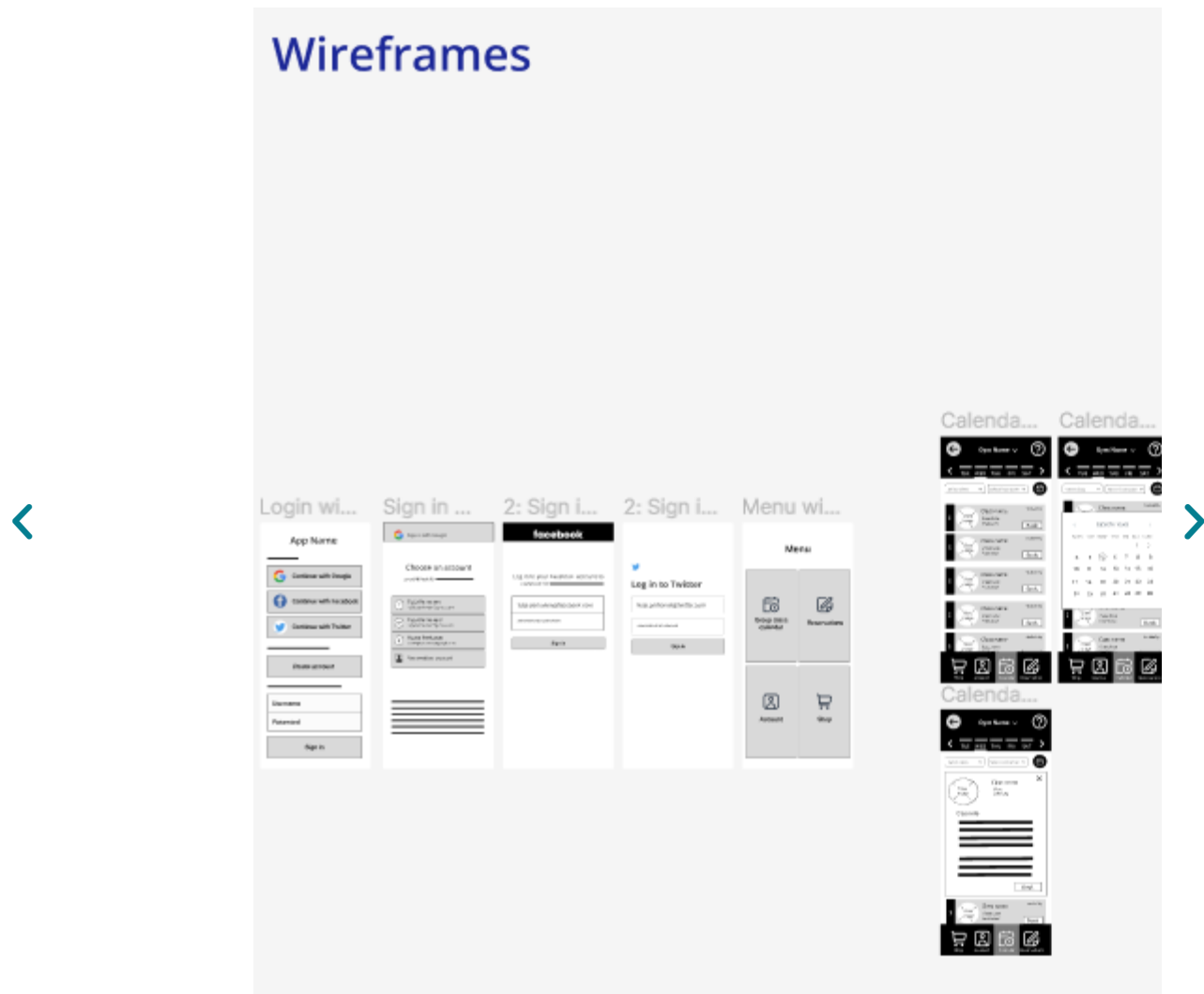
Page Width

Working within an Agile framework, I iteratively developed the app, with regular sprints and feedback loops. This allowed me to refine the app based on continuous testing and user feedback, ensuring that I were building a product that truly met the needs of gym users.

8. UI/UX Design and Usability Testing

On the design side, I created the User Interface (UI) for the Gym Booking App. I applied Gestalt Principles to guide the visual design, ensuring the app was intuitive and easy to navigate.

To validate the design, I created wireframes and app's interactive prototype using Figma. These prototypes were then tested with users to gather feedback on usability and identify areas for improvement. The iterative process of usability testing helped me refine the app's user experience, ensuring that the final product was both functional and user-friendly.



After the first sprint, I also conducted a Heuristic Evaluation to identify potential usability issues early in the design process.

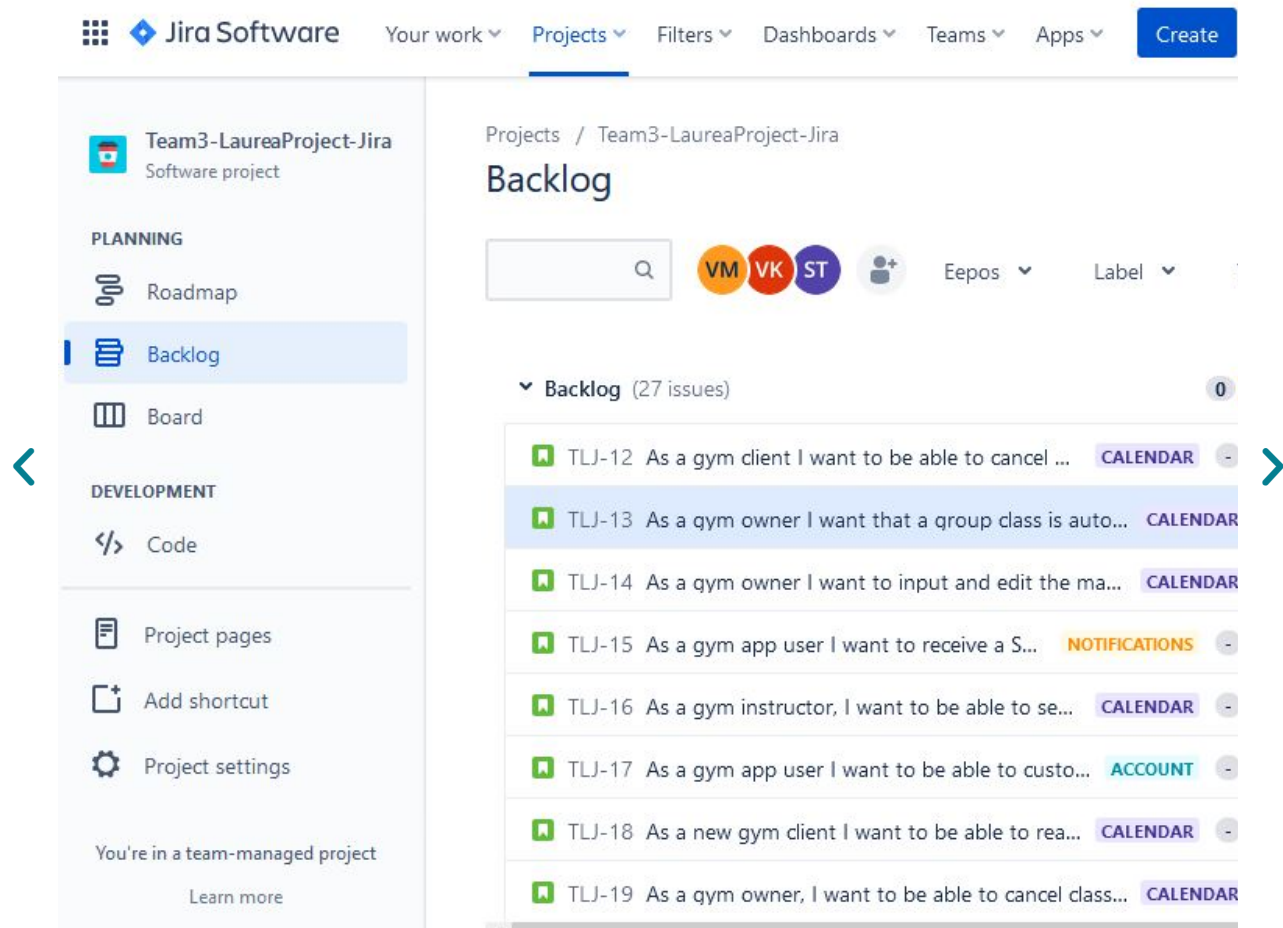
Nielsen Usability Heuristics				
Problem (How you describe the usability problem?)	Heuristic violated (What heuristic is violated and how?)	Problem severity	Improvement suggestion (How UI design should be improved?)	Wireframes updated (Did you already update wireframes accordingly?)
Text links are invisible: text link color is indistinguishable from regular text	Consistency and standards: External consistency should be established for improving app usability. The user should intuitively recognize links and see straight that text is clickable.	Medium	Adding visible links to text in following pages: - Sign into class: Queue -> You can see all your reservations in the Reservation page. - Sign into class: Successful -> You can see all your reservations in the Reservation page.	Updated
Create account-button is between the sign in-fields.	Match between system and the real world: Create account field/button is typically under the Sign in-fields.	Medium	Create account button should be moved under all the sign in-fields (Google etc. and sign in with existing account).	Updated
Cancel/back-buttons are missing on the Calendar month pop-up window and the Booking confirmation, Booking details pop-up forms	User control and freedom: User should always have a clear exit to leave unwanted action.	High	Back or cancel buttons should be added to page: - Calendar month (for the user to exit the monthly calendar). - Booking confirmation form (the user has a possibility to close the pop-up form) - Booking details form (the user has a possibility to close the pop-up form)	Updated
There is inconsistency with the words used throughout the app: reservation and booking are mixed. There is the rarely used "reservation" word that is used above the more common word "booking".	Consistency and standards: User should not wonder if different words mean the same thing. The user should see a word that is more familiar to him.	Low	Use either book & bookings or reserve & reservations throughout the pages, but do not mix the words. Book & bookings words should be chosen in the app. Reserve & reservations words should be replaced with book & bookings in the app.	Updated
There is unnecessary information on calendar page: class room, class ratings.	Aesthetic and minimalist design: Interface should contain only essential information.	Low	Some class details, such as room and ratings could be hidden in the main calendar view and visible only in class details-pop up window.	Updated
There is no help button if the user has problems with using the application.	Help and documentation: Help and documentation content should be easy to search and focused on the user's task.	Low	There could be a question mark-icon in the pages where user can access an application user guide.	Updated

9. Atlassian Jira and Confluence

I used both Atlassian Jira and Atlassian Confluence to streamline the development process and ensure effective collaboration throughout the project. Jira helped me manage the app's backlog, organize tasks, and track progress. I created user stories, prioritized tasks, and tracked development through sprints, ensuring smooth execution from start to finish.

On the other hand, Confluence served as the central documentation hub. I used it to store and share key project information, such as the database schema, wireframes, technical specifications, and user stories. It allowed me to keep the team aligned by documenting requirements, sprint goals, and feature descriptions, while tracking decisions and gathering feedback.

By integrating Jira and Confluence, I ensured seamless communication and a structured workflow, making it easier to manage development tasks, track progress, and maintain comprehensive project documentation.



Outcome and Reflection

At the conclusion of the Gym Booking App project, I gained a comprehensive understanding of various software development methods and tools. I learned how to apply Agile principles to a real-world project, as well as how to create technical documentation, design databases, and integrate user-centered design practices into the development process. More importantly, I developed a holistic approach to managing software projects, balancing both technical requirements and user needs.

The project not only deepened my technical skills but also reinforced the importance of collaboration, user feedback, and continuous iteration in the software development lifecycle.

You can explore the interactive prototype of the Gym Booking App created by my team in [Figma](#).

Challenges

During the development of the Gym Booking App, I faced several challenges. Adapting to Agile methodologies, especially Scrum, was initially difficult as I learned to manage sprints, prioritize tasks, and handle shifting requirements. Collaborating with a cross-functional team also presented coordination challenges, requiring clear communication and alignment on goals. Balancing technical functionality with user-centered design was challenging, as was incorporating user feedback while maintaining a cohesive vision. Additionally, keeping software documentation up to date while working on development and design required strong organizational skills. Despite these challenges, they provided valuable learning experiences and helped me develop essential problem-solving and communication skills.

Tools

Miro, Figma, Atlassian Confluence and Jira

The Gym Booking App's presentation:





 GitHub

 LinkedIn



viktoriya.mikhaylova@hotmail.com

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